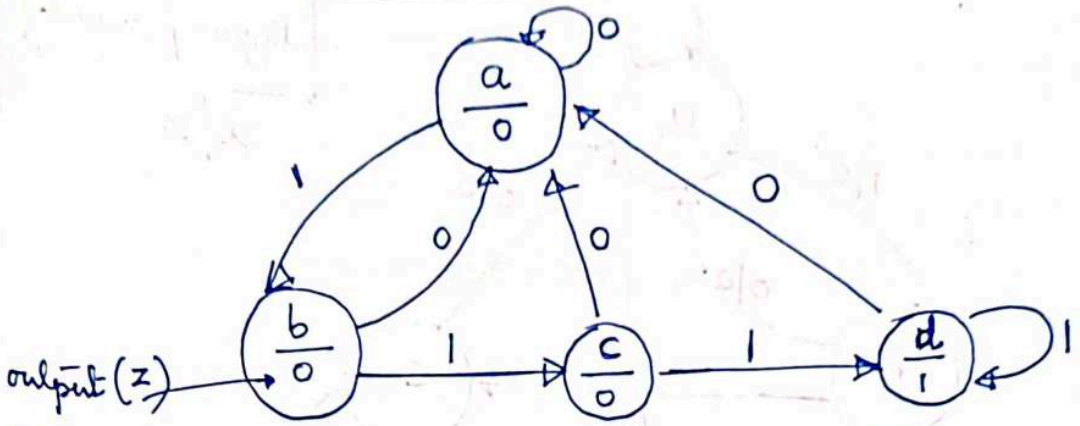
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Example of state diagram of Moore model :-



In the above diagram,

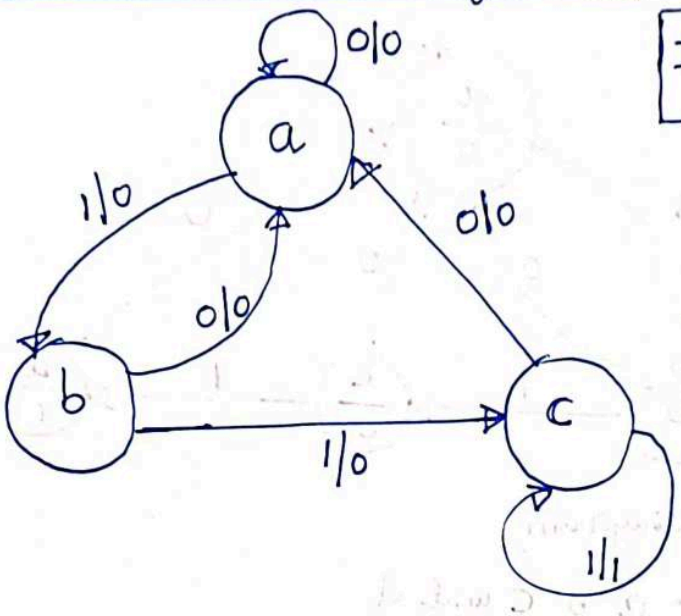
1) 4 states: a, b, c and d
2) Input: x
3) output: Z

* State Table :

present state	Next state		output
	I/P	I/P	
	$x=0$	$x=1$	Z
a	a	b	0
b	a	c	0
c	a	d	0
d	a	d	1

Example of state diagram of Melay model :—

Input	output
x	z

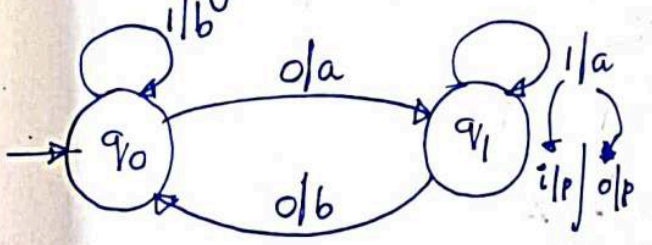
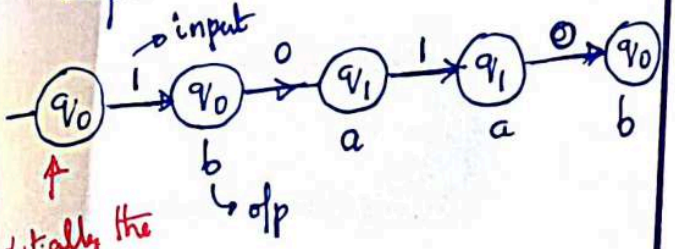
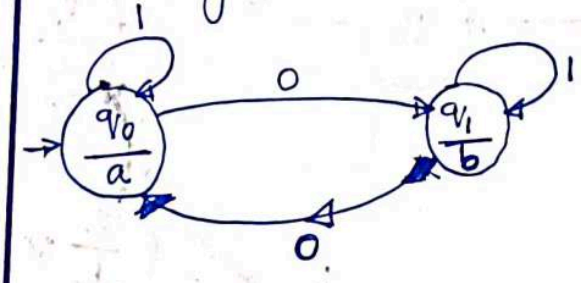
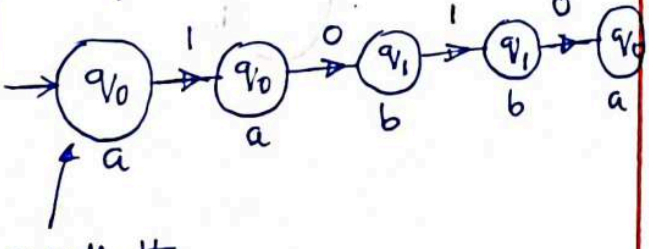


* State Table :

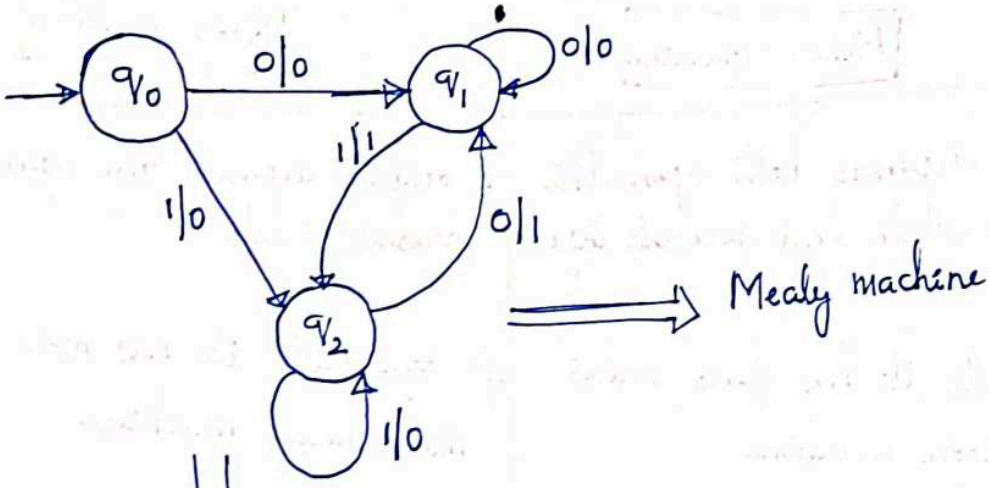
present state	Next state		output (z)	
	x=0	x=1	x=0	x=1
a	a	b	0	0
b	a	c	0	0
c	a	c	0	1

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* Differences between the Melay and Moore Machine:

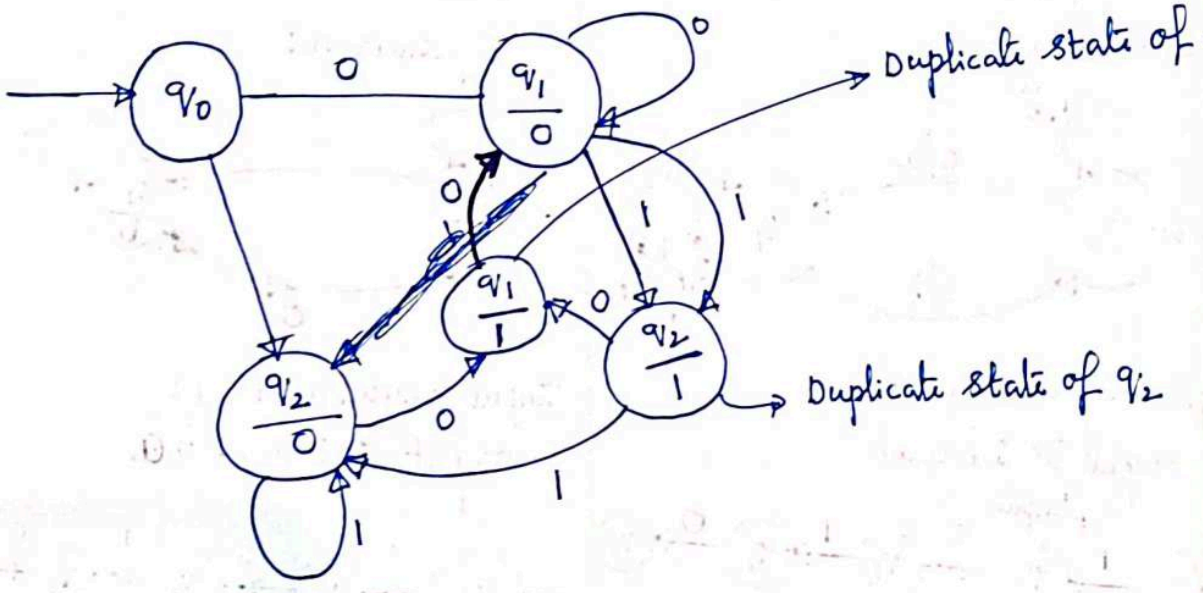
Melay Machine	Moore Machine
1) output depends both upon the present state and present input	1) output depends only upon the present state
2) Generally it has fews states than Moore machine	2) Generally it has more states than mealy machine
3) Example: State diagram  Input sequence: 1010 output is: baab  Initially the state is q_0	3) Example: State diagram:  Input sequence: 1010 output is: aabbab  Intially the state is a
4) Input string of length 'n' → output length is also 'n'	4) Input string length 'n' → output string length 'n+1'
5) output is asynchronous	5) output is Synchronous with clock.

* Mealy to Moore conversion with Example :



Mealy machine

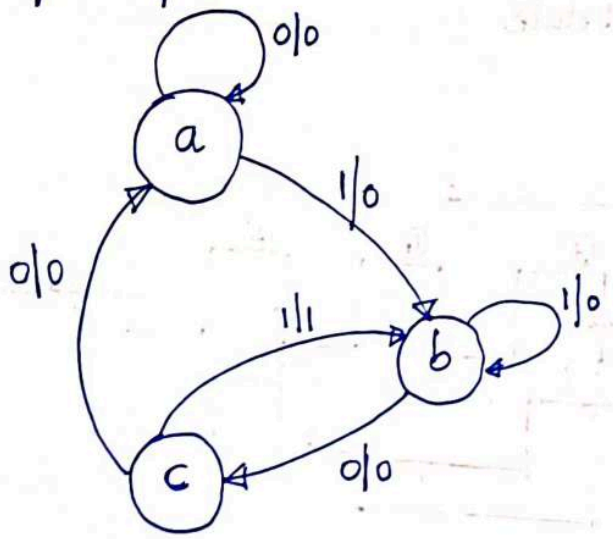
Moore machine



* Melay state diagram for the serial input sequence 101

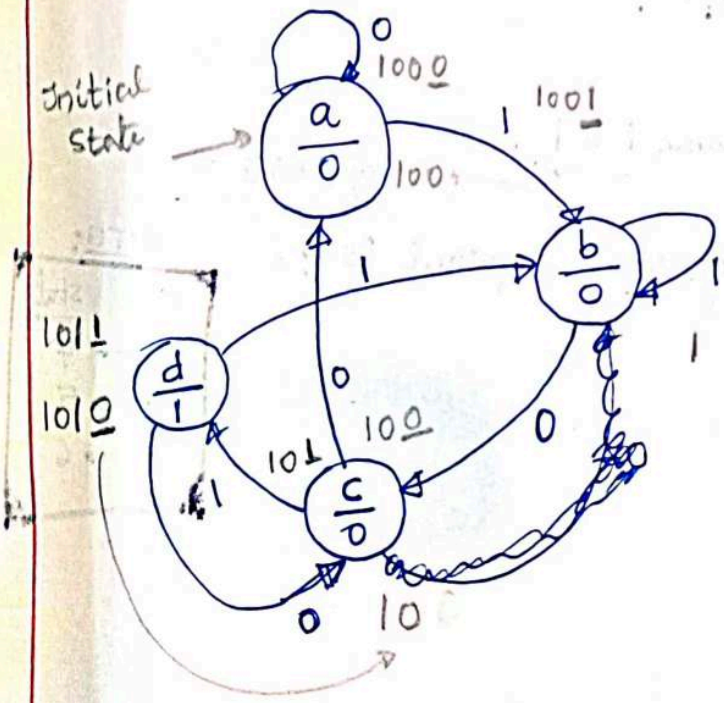
Input sequence: 101

I/p → 101
↓
1 → o/p



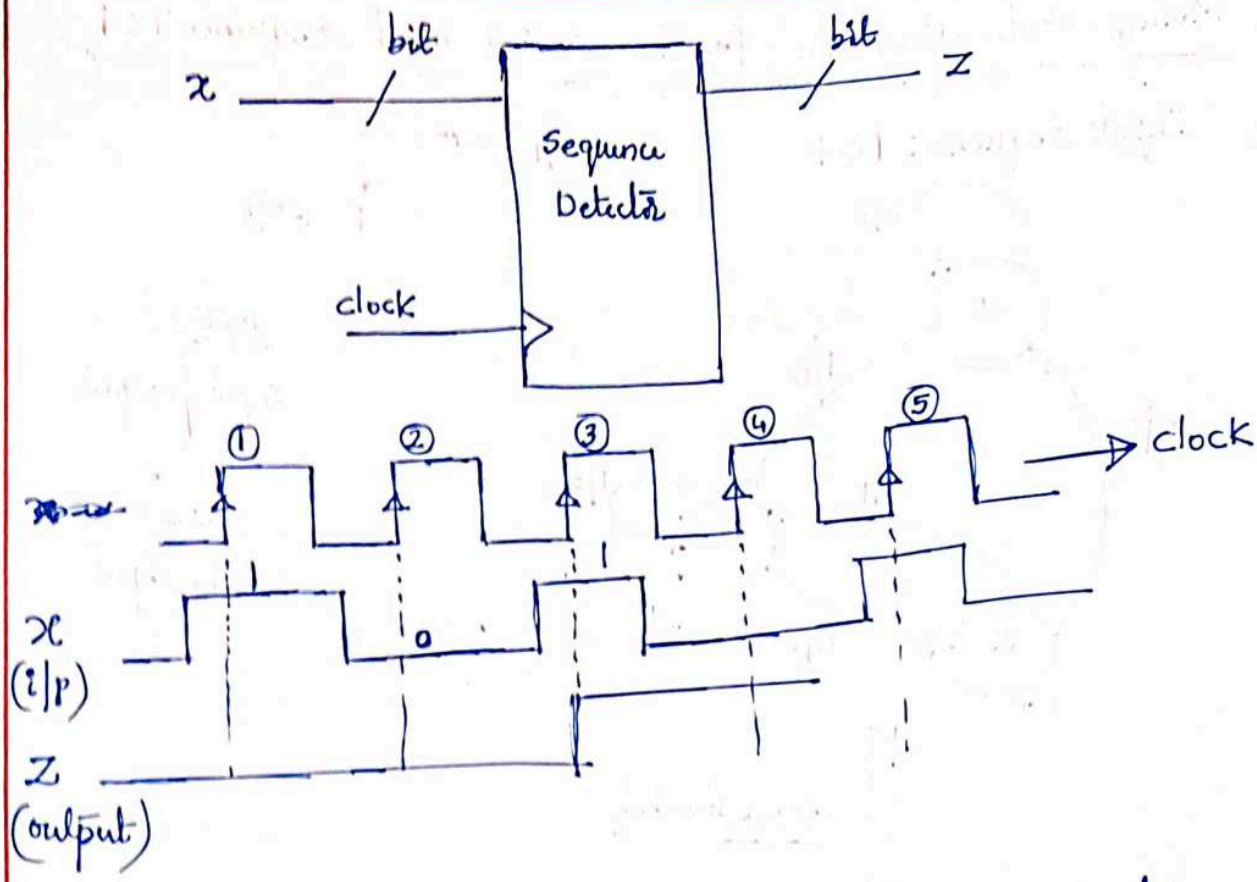
NOTE:
Input / output
↓
Depending on the input

Moore model



1011
↓
1
↓
10000
↓
0

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NOTE: Sequence detector is used in Digital communication.

Moore Machine: For the Sequence, 1 0 1 → No. of bits 3

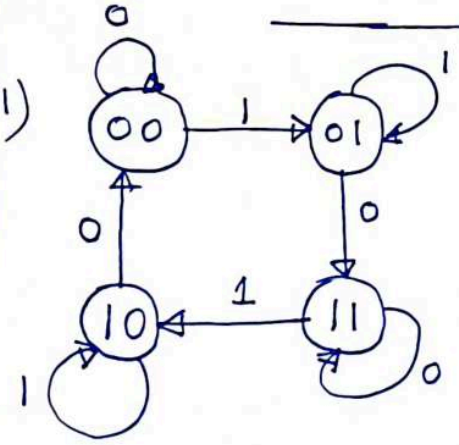
then the no. of states required is 4.

NOTE:

Bits	states
3	→ 4
4	→ 5
5	→ 6

From FSM ⟶ Digital circuit

1)



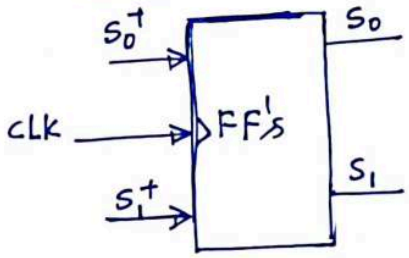
Input = x

State Variables = $S_0 S_1$ → Each is represented by two bits.

2)

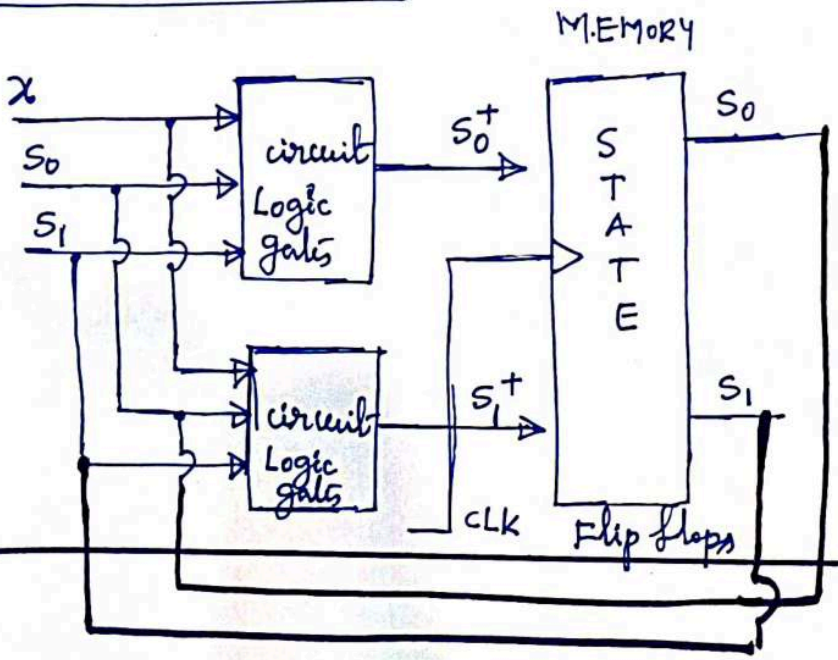
x	S_0	S_1	S_0^+	S_1^+
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	1	0
1	1	1	1	0

NOTE:



After clock transition, Future state of the machine becomes the current state of the machine.

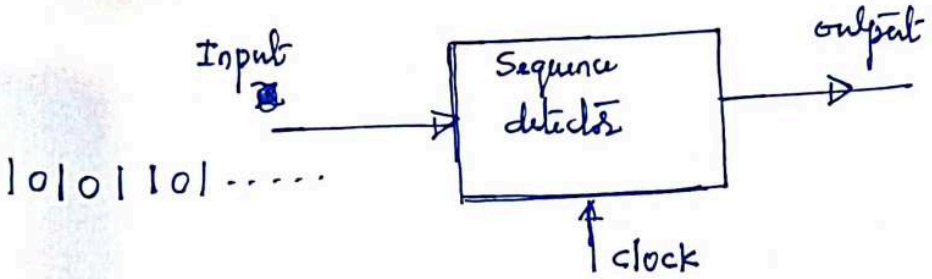
3)



* Synthesis: → state Diagram ⟶ circuit

* Analysis: → circuit ⟶ state diagram

* Sequence Detector | Recogniser :



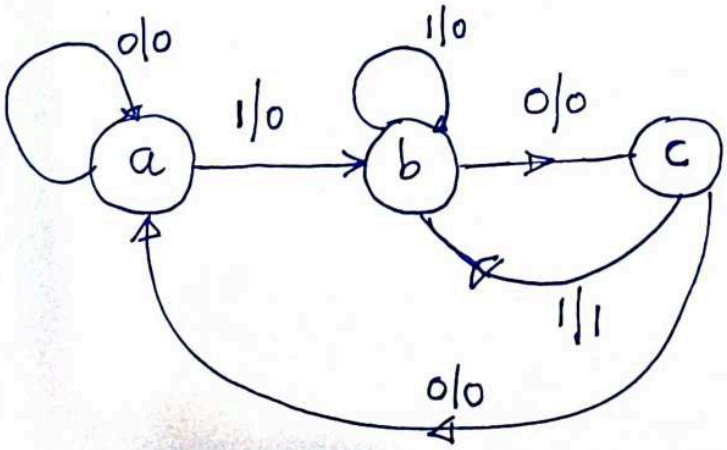
→ whenever the particular sequence is detected then the output becomes 1

Example: consider sequence, 1 0 1

The Initial state is (a)

(b) is the state which gets first one in the sequence

(c) is the state which gets 1 0



i/p = 1 0 1 0 1